

GEN AI Bootcamp

Day 1 Homework – Multimodal AI

Problem Statement

Imagine you are building an AI system for an e-commerce platform like Amazon or Flipkart.

The system should be able to **understand a product using image**.

1. Inputs

- Product image



A seller uploads a product image (e.g., a red dress) to the e-commerce platform. The AI system analyzes the image to identify the product category, extract key attributes, and generate a product description automatically.

2. Outputs

After analyzing the uploaded product image, the AI system generates the following outputs:

- **Product category**
Women's Clothing → Dress
- **Product Attributes**
Color: Red
Product Type: Dress
- **Product Description**
Elegant red casual dress with a comfortable fit and modern design. Suitable for everyday wear and special occasions.

3. Model Selection

To understand product images and generate text, the system uses a Multimodal AI Model.

Chosen Model: Florence-2

Why Florence-2?

Understands product images.

Identifies product category and attributes.

Generates product descriptions.

4. System Design (High-Level)

The system automatically converts a product image into useful product information. After a seller uploads an image, the AI model analyzes it and generates the product category, attributes, and description.

Architecture Diagram



5. Evaluation

I will use the following metrics to check whether my model is performing well:

- **Accuracy:** To verify if the product category is predicted correctly.
- **Description Quality:** To check whether the generated description is relevant, clear, and meaningful.
- **Attribute Accuracy:** To verify if attributes such as color, product type, and style are identified correctly.

If the model achieves high accuracy and generates relevant descriptions and attributes, I can conclude that the model is performing well.

6. Business Value

This system helps sellers save time by automatically generating product information from an image, such as a red dress. It reduces manual effort, improves product listings, and enhances the shopping experience for customers.

Real-world company using multimodal AI

Amazon

Amazon uses multimodal AI to analyze product images and text together. It helps automatically categorize products, generate product information, improve search results, and recommend similar products, making online shopping more efficient.